

Model-Based Meta-Analysis of Oral Acetaminophen Pharmacokinetics

A Detailed, Step-by-Step Analyst's Manual

Every step explained from first principles • every abbreviation spelled out • code walked through line by line

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Reader	An analyst new to this project — assumes no prior knowledge
Covers	Steps 0-4: setup, literature review, extraction, dataset, model fit
Core tool	admrx2 0.2.0 (built on nlmixr2 and rxode2)

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0. How to Read This Document

This manual is written for an analyst who has just joined the project and knows nothing about it yet. Every technical term is explained the first time it appears, and every abbreviation is written out in full. Where the project involves running code, the code is shown and then explained one line at a time in a two-column table: the left column is the exact code, the right column is what it means in ordinary language.

You do not need to read it all at once. Section 1 is a dictionary you can return to. Sections 2-3 give the big picture. Sections 4-8 explain each step of the actual work. Boxes flagged **Key idea** capture the single most important point of a section; boxes flagged **Note** record honest caveats and practical warnings.

1. Abbreviations and Plain-Language Glossary

Every abbreviation and specialist word used anywhere in this project, expanded and explained in everyday terms.

1.1 Abbreviations

Short form	Full form	In plain words
PK	Pharmacokinetics	The study of how the body absorbs, distributes and removes a drug over time – "what the body does to the drug"; "what the drug does to the body".
PD	Pharmacodynamics	The drug's effect on the body (not the focus here, but admixr2 supports it).
MBMA	Model-Based Meta-Analysis	Pooling results from many published studies by fitting one mechanistic model to their summary data.
IR	Immediate-Release	An ordinary tablet/capsule that releases all its drug at once (as opposed to slow-release).
ODE	Ordinary Differential Equation	A maths equation describing a rate of change – here, how fast drug amount in a body changes.
NLME	Non-Linear Mixed-Effects	The statistical model family for population PK: a typical curve plus person-to-person variation.
IIV	Inter-Individual Variability	How much subjects differ from each other (also called between-subject variability).
C _{max}	Maximum concentration	The highest drug level in blood after a dose.
T _{max}	Time of maximum	How long after dosing C _{max} occurs.
AUC	Area Under the Curve	Total drug exposure – the area under the concentration-vs-time graph.
t _{1/2}	Half-life	Time for the blood level to fall by half.
k _a	Absorption rate constant	How fast drug moves from the gut into the blood.
CL	Clearance	Volume of blood cleared of drug per hour (how fast the body removes it).
V	Volume of distribution	An apparent volume linking amount of drug in the body to its blood concentration.
F	Bioavailability	Fraction of an oral dose that actually reaches the bloodstream (0-1).
CL/F, V/F	Apparent clearance / volume	Clearance and volume divided by F – the only versions estimable from oral-only data.
E	Mean vector	The list of average concentrations at each sampling time for a study (an admixr2 input).
V (matrix)	Covariance / variance	How those means scatter – their spread and correlation (an admixr2 input).
FO	First-Order	A fast approximate estimation method (an admixr2 estimator).
MC	Monte Carlo	Estimation by random sampling – the main admixr2 estimator.
GH	Gauss-Hermite	A noise-free numerical alternative to Monte Carlo.
IRMC	Iterative Reweighting Monte Carlo	A faster Monte Carlo variant for hard problems.
AIC	Akaike Information Criterion	A score for comparing models; lower is better, and it penalises needless complexity.
-2LL	Minus two log-likelihood	The number the fitting tries to minimise; smaller means the model fits the data better.
PRISMA	Preferred Reporting Items for Systematic Reviews and Meta-Analyses	A standard checklist for reporting how studies were found and selected.
PICO	Population, Intervention, Comparator, Outcome	A framework for writing precise study-selection criteria.
NCA	Non-Compartmental Analysis	A model-free way papers summarise PK (gives C _{max} , T _{max} , AUC directly).
CSV	Comma-Separated Values	A plain-text spreadsheet file.

Short form	Full form	In plain words
API	Application Programming Interface	The set of function names and arguments a software package exposes for you to call.

2. The Big Picture: What Are We Actually Doing?

Imagine forty different research teams over fifty years each gave the same drug, oral acetaminophen, to a handful of healthy volunteers and measured the blood levels over time. Each team published a small graph and a few summary numbers, then moved on. Individually, each study is too small to say much. Together they contain a great deal of information — if only we could combine them properly.

The problem: we almost never get the original patient-by-patient measurements. We only get the **published summaries** — averages, standard deviations, and the number of subjects. A **model-based meta-analysis (MBMA)** is a technique for fitting a single pharmacological model to exactly that kind of summarised evidence.

Key idea — Instead of averaging reported numbers, we fit a mechanistic model of drug absorption and elimination directly to the summary statistics of many studies at once. The tool that makes this possible in R is the **admixr2** package.

2.1 Why acetaminophen, healthy adults, immediate-release?

Acetaminophen (called paracetamol outside the United States) is one of the most-studied oral drugs in the world, so there is abundant published data. Restricting to **healthy adults** and **immediate-release** tablets removes complications (disease, unusual formulations) that would otherwise muddy the absorption picture. We deliberately keep the analysis clean first; dose, food, age and sex are then tested as explanatory factors afterwards.

3. The Software Stack (What Each Tool Is)

Three R packages do the modelling, each building on the one below it. Think of them as three floors of a building.

Package	What the name means	What it does for us
rxode2	"R" + "ODE" (ordinary differential equations)	The ground floor. It solves the equations that describe drug moving between body compartments.
nlmixr2	Non-Linear Mixed-effects in R	The middle floor. It adds the population statistics layer (a typical patient plus variability).
admixr2	Aggregate-Data Mixed-effects in R	The top floor, and the heart of this project. It lets the nlmixr2 models be fitted to summary data.

Note — These packages compile C code when first installed, so a compiler toolchain is required: **Rtools** on Windows or the **Xcode command-line tools** on macOS. This is why the very first install can take several minutes.

4. Step 0 — Setting Up the Environment (00_setup.R)

Before any science, we make the project **reproducible** — meaning anyone can re-run it and get identical results. Reproducibility comes from pinning down two things: which software versions we used, and where every file lives. The setup script does both.

4.1 Line-by-line walkthrough

R code	What it means in plain English
<pre>pkgs <- c("rxode2", "nlmixr2", "admixr2", "tidyverse", "readxl", "here")</pre>	Make a list of the six packages the project needs. 'tidyverse' is a bundle for data handling and plotting; 'readxl' reads Excel files; 'here' builds file paths relative to the project so code works on any computer.

R code	What it means in plain English
<code>missing <- setdiff(pkgs, rownames(installed.packages()))</code>	Compare the needed list against what is already installed, and keep only the ones that are missing.
<code>if (length(missing)) install.packages(missing)</code>	If anything is missing, download and install it. Written this way the script is safe to run again and again.
<code>loaded <- vapply(pkgs, require, ...) if (any(!loaded)) stop(...)</code>	Load each package into the session. If any fails to load, halt immediately with a clear message rather than continuing in a broken state.
<code>paths <- list(data_raw=here::here("data", "raw"), ...)</code>	Build one central list of folder locations. Later scripts refer to <code>paths\$data_raw</code> etc., so folder names live in exactly one place.
<code>for (p in paths) dir.create(p, showWarnings=FALSE, recursive=TRUE)</code>	Create any missing folders so nothing downstream breaks.
<code>writeLines(capture.output(sessionInfo()), ".../sessionInfo.txt")</code>	Write out the exact version of R and every package to a text file — the permanent record of what produced the results.
<code>set.seed(20260711)</code>	Fix the random-number seed. The Monte Carlo estimator later uses randomness; a fixed seed makes it repeatable.

Key idea — Run this file as a whole (RStudio's **Source** button), never line by line. Each line depends on the ones above it — running them out of order is what caused the earlier "object 'pkgs' not found" error.

5. Step 1 — Systematic Literature Review

A **systematic literature review** is a transparent, rule-based search for all the studies that qualify, decided by criteria fixed *before* looking at results so the choice cannot be biased after the fact.

5.1 How the selection rules were written (PICO)

PICO stands for **P**opulation, **I**ntervention, **C**omparator, **O**utcome — a standard way to state precisely who and what a review includes:

Letter	Stands for	Our rule, and the reason
P	Population	Healthy adults only. Disease and frailty change how fast the body clears the drug, which would add noise we can't control.
I	Intervention	Oral immediate-release tablet/capsule. Effervescent, slow-release and liquid forms alter absorption & metabolism.
C	Comparator	None (this is descriptive). Intravenous arms are used only to work out bioavailability F.
O	Outcome	A study must report usable PK numbers (C _{max} , T _{max} , AUC, half-life, CL/F, V/F) and/or a mean concentration-time curve.

5.2 What PRISMA is, and the result

PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) is an internationally agreed way to report exactly how many records were found, screened, and kept or dropped, so the process is auditable. Applying our rules: 18 candidate records were catalogued, **9 were included**, and 9 were excluded (3 of those kept only as background references). The nine cover doses of 500, 650, 1000 and 2000 mg, single and repeated dosing, tablets and capsules, both sexes, fasted and fed, and both older (1977-1983) and modern (2003-2018) studies. That spread is what lets us estimate the model and later test a dose effect.

Note — This was a rapid web-based scoping search, not yet a formal indexed database search of PubMed and Embase (two major medical literature databases). Before publishing, it should be repeated formally with recorded hit counts and a second reviewer. The two deliverables produced — a written review and a colour-coded screening spreadsheet with a reason for every paper — are built to accept those final numbers.

6. Step 2 — Extracting the Data (02_data_extraction.R)

This step turns nine papers into a single tidy table: one row per **study-arm-dose** (an "arm" is one treatment group within a study), one column per standardised measurement.

6.1 The two rules that protect the analysis

Rule 1: one unit system

Different papers report the same quantity in different units. If we mix them, the model fits nonsense. We convert everything on entry to a single system: concentration in **mg/L**, exposure (AUC) in **mg·h/L**, time in **hours**, clearance in **L/h**.

Two worked conversions that come up constantly:

- **Concentration:** 1 mg/L is exactly the same as 1 µg/mL (microgram per millilitre). So a paper reporting "9.46 µg/mL" needs no numeric change — only the label changes to 9.46 mg/L.
- **Clearance:** a paper reporting "350 mL/min" is converted by multiplying by 0.06 to get 21 L/h (because 60 min/hour ÷ 1000 mL/L = 0.06).

Rule 2: never invent a number

Every row carries a flag called `data_status` with one of two values: **extracted** means the value was actually read from the paper and its source recorded; **to_extract** means the paper qualifies but the number still has to be transcribed from its printed table or figure. Only verified numbers are filled in. Nothing is guessed. Every row also names its `source_ref`.

6.2 What the script does

R code	What it means in plain English
<code>raw <- read_csv("../pk_summary_extraction.csv")</code>	Read the hand-entered extraction sheet into R.
<code>stopifnot(all(required %in% names(raw)))</code>	Confirm the essential columns exist; halt if any are missing.
<code>filter(dose_mg<=0 cmax<=0 f_abs<0 f_abs>1)</code>	Flag impossible values — a non-positive dose or concentration, or a bioavailability outside 0-1 — before they can corrupt the model.
<code>count(data_status)</code>	Report how many rows are done versus still pending, so progress is visible.
<code>write_csv(clean, "../pk_summary_clean.csv")</code>	Save the checked, sorted table for the next step.

7. Step 3 — Building the MBMA Dataset (03_build_dataset.R)

The model is fitted to the **concentration-time profiles** (the curves), not just the headline numbers. This step assembles those curves into the exact shape the fitting engine expects.

7.1 The single most important concept: aggregate data carries N and SD

In a meta-analysis the **study** is the unit of observation. A study should count more if its average was measured precisely. Precision depends on two things: the spread of the measurements (standard deviation, SD) and the number of subjects (N). So every point on every curve carries three numbers, not one:

Column	Meaning	Why it matters
<code>dv_mean</code>	Mean concentration at that time	The actual observation to be matched by the model.
<code>dv_sd</code>	Standard deviation at that time	How scattered subjects were — the model's uncertainty at that point.
<code>n</code>	Number of subjects	A 40-subject study is more trustworthy than a 6-subject one.

The script also computes the **standard error of the mean**, SD divided by the square root of N (written $SD/\sqrt{N}$). This is the precision of the study's *average*, and it is what an aggregate-data model weights each study by.

7.2 Where the curves come from: digitising figures

Old papers print a graph, not a data table. To recover the numbers we use a free tool called **WebPlotDigitizer**: you load the figure image, tell the tool where the axis tick marks are (calibration), then click along the printed mean curve. The tool converts each click back into a (time, concentration) pair. Error bars give the SD. These pairs are pasted into `data/raw/pk_profiles.csv`, and the script validates them and writes `mbma_dataset.csv`.

8. Step 4 — Fitting the PK Model (04_fit_model.R)

This is the scientific core. We define a mechanistic model of how acetaminophen moves through the body and fit it to the aggregate dataset with `admixr2`. This section is deliberately the most detailed.

8.1 First, how I found out how `admixr2` works

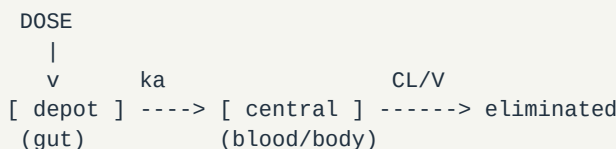
`admixr2` is a young, specialised package (version 0.2.0). To use it correctly I did **not** guess the function names — guessing an interface is how silent errors creep in. Instead I looked up the package's official documentation and read its worked example. Here is exactly what that told me, so you can verify it yourself:

Source consulted	What it confirmed
CRAN package page	<code>admixr2</code> fits PK/PD models to "aggregate-level data (mean vector and covariance matrix per study)" and offers four estimators
Package function index	The fitting is done through <code>nlmixr2</code> using estimator methods named <code>adfo</code> , <code>admcmc</code> , <code>adgh</code> , <code>adirmc</code> , each with a matching <code>ad</code> function
"Getting started" vignette	The exact recipe: define a normal <code>nlmixr2</code> model, then supply each study as a list with five elements (<code>E</code> , <code>V</code> , <code>n</code> , <code>times</code> , <code>events</code>)

Note — A **vignette** is a tutorial document shipped inside an R package. Reading the vignette is the reliable way to learn a package's real interface (its API — the set of functions and arguments it expects), rather than assuming. Everything in the fit script below matches that official example.

8.2 The structural model, explained as a picture

We start with the simplest sensible model for an oral drug: a **one-compartment model with first-order absorption**. "One-compartment" means we treat the whole body's blood and tissues as a single well-mixed tank. "First-order absorption" means the drug moves from the gut into the blood at a rate proportional to how much is still in the gut. In pictures:



Drug is swallowed into the **depot** (the gut). It is absorbed into the **central** compartment (blood and body) at rate constant **ka**. From there it is removed (metabolised/excreted) in proportion to its clearance **CL** and inversely to the volume **V**. The concentration we compare against data is amount in the central compartment divided by V.

The model in code, line by line

R code	What it means in plain English
<code>d/dt(depot) = -ka * depot</code>	Rate of change of drug amount in the gut = it only leaves, at speed <code>ka</code> times the amount present (so it empties fastest when fullest).

R code	What it means in plain English
$d/dt(\text{central}) = ka \cdot \text{depot} - (CL/V) \cdot \text{central}$	Rate of change in the blood = what arrives from the gut ($ka \cdot \text{depot}$) minus what is cleared ($(CL/V) \cdot \text{central}$).
$cp = \text{central} / V$	Predicted plasma concentration = amount in the central tank divided by its volume. This is what we match to the measured curve.

8.3 Two modelling decisions, and why

Decision 1: we estimate CL/F and V/F, not CL and V

F is bioavailability — the fraction of a swallowed dose that actually reaches the blood (some is lost to the gut wall and liver first). With **oral data only**, the mathematics cannot separate clearance CL from bioavailability F: they always appear together as the ratio CL/F. The same is true for volume, giving V/F. These ratios are called **apparent** clearance and volume. This is a fundamental limit of the data, not a shortcut — you would need an intravenous arm to pin down F on its own.

Decision 2: mu-referencing

Each parameter is written on the **logarithmic scale** as $\exp(\text{theta} + \text{eta})$. Here **theta** is the typical (population-average) value and **eta** is an individual's deviation from it. Writing it as $\exp(\dots)$ guarantees the parameter stays positive (a negative clearance is meaningless) and makes the population-average and the individual-variation add together neatly on the log scale. This particular pattern is called **mu-referencing**, and **admixr2** specifically requires it because it lets the package compute exact mathematical gradients, which makes fitting faster and more stable.

R code	What it means in plain English
<code>tka <- log(1.5)</code>	Typical absorption rate ka starts at 1.5 per hour (log-scale starting guess).
<code>tcl <- log(21)</code>	Typical apparent clearance CL/F starts at 21 L/h — equal to ~350 mL/min, the value Rawlins measured in 1977.
<code>tv <- log(60)</code>	Typical apparent volume V/F starts at 60 L, a literature-typical figure for acetaminophen.
<code>eta.cl ~ 0.09</code>	Allow clearance to vary between studies; 0.09 on the log scale is roughly 30% variability as a starting point.
<code>cp ~ prop(prop.sd)</code>	Assume the measurement error grows in proportion to the concentration (a 'proportional' error model).

8.4 Handing the data to admixr2: the five ingredients

For each study arm the script builds the five objects **admixr2** asks for. Using a concrete imaginary study (1000 mg dose, sampled at 0.5, 1 and 2 hours) to make it real:

Ingredient	What it is	Example for one study
E	Mean concentration at each time (the 'mean vector')	c(6.2, 9.1, 7.3) mg/L
V	Variance of each mean (spread, SD squared)	c(1.4, 1.8, 1.6)^2
n	Number of subjects in the arm	12
times	The sampling times	c(0.5, 1, 2) hours
ev	The dosing event	one 1000 mg oral dose

The script loops over every study and packs these into a named list called **studies**. Because published papers almost never report how concentrations at different times correlate with each other, we supply **V** as a simple list of variances (the diagonal) rather than a full matrix — a standard, slightly conservative choice.

R code	What it means in plain English
<pre>studies[[sid]] <- list(E = d\$dv_mean, V = d\$dv_sd^2, n = d\$n[1], times = d\$time_h, ev = et(amt=d\$dose_mg[1]))</pre>	For study 'sid', bundle its mean curve (E), the variances (V = SD squared), the subject count (n), the sampling times, and a single oral dose event (ev). et() is rxode2's helper for building a dosing schedule.
<pre>fit <- nlmixr2(oral_1cmt, admData(), est="admc", control=admControl(studies=studies, ...))</pre>	Fit the model. 'oral_1cmt' is our model; admData() is a placeholder table admixr2 requires; est="admc" chooses the Monte Carlo estimator; the studies list is passed inside admControl().

8.5 Choosing an estimator (the 'est=' setting)

An **estimator** is the numerical method that searches for the parameter values making the model best match the data. admixr2 offers four, trading speed against exactness. You pick one with the est= argument:

Estimator	est =	In plain words	When to use
First-Order	adfo	Fast, approximate	Quick first look, rough starting values
Monte Carlo	admc	Uses random sampling; accurate	The default workhorse — what we use
Gauss-Hermite	adgh	Noise-free numerical version	Models with only a few varying parameters
Iterative Reweighting MC	adirmc	A faster Monte Carlo for hard cases	Complex or slow models, poor starting values

8.6 What the fit produces

After fitting, the script:

- Prints the fit and back-transforms the log-scale estimates into readable numbers: the typical **ka** (per hour), **CL/F** (L/h) and **V/F** (L).
- Reports two fit-quality numbers: the **objective function** (minus-two-log-likelihood, -2LL — smaller is a better fit) and the **AIC** (Akaike Information Criterion — used to compare rival models, penalising extra complexity).
- Saves the fitted model object to models/fit_oral_1cmt.rds and the parameter table to results/params_oral_1cmt.csv.

Note — Because publications give only per-time spread and not the cross-time covariance, V is supplied as a diagonal variance vector. This is standard practice and mildly conservative (it can slightly widen the uncertainty on parameters); it should be stated plainly in the final write-up.

9. What Comes Next

- Two-compartment comparison:** acetaminophen can show an early rapid-distribution phase; we will fit a two-tank version and compare it to the one-tank model by AIC.
- Step 5 — Covariate analysis:** test whether dose, formulation (tablet vs capsule), food, age and sex explain part of the between-study differences.
- Step 6 — Plots:** observed-versus-predicted curves and goodness-of-fit figures for publication.
- Publish** the whole reproducible workflow on GitHub.

10. Appendix: The Research Trail

For full transparency, these are the external sources consulted while building the model step, and what each provided. They can all be re-checked.

Purpose	Source	What it gave
Confirm admixr2 is right for the job	CRAN package description	Confirmed it fits PK models to aggregate mean+covariance data — ex
Confirm the drug's PK values	Rawlins 1977; Divoll 1982; 2016 bioequivalence study	Clearance ~350 mL/min, dose-dependent bioavailability, Cmax/Tmax ranges
Learn the exact admixr2 interface	admixr2 'Getting started' vignette + function Title	E/V/n/times/ev study format, the four estimators, and the nlmixr2 fitting o
Confirm the modelling layers	nlmixr2 documentation	That nlmixr2/rxode2 provide the model and ODE-solving engine admixr2 sits

Companion files: *README.md*; *docs/01_systematic_literature_review.md*; *docs/literature_screening_log.xlsx* (one reason per paper, with clickable source links); and *scripts/00_setup.R* through *04_fit_model.R*. This manual documents the project through Step 4.